BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 3485 / 3486

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 1022 (جماعت دہم) — سیکنڈری پارٹ II — Session 2018-20 to 2020-22 — کیمسٹری (معروضی) — گروپ پہلا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 3485

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Just above the Earth's Surface is.

(A) (B)

(C) (D)

Q2. Temporary hardness is because of

(A) (B)

(C) (D)

Q3. Which one of the following gas is used to destroy harmful bacteria in water?

(A) (B)

(C) (D)

Q4. Crude oil is heated in furnace upto

(A) (B)

(C) (D)

Q5. For reaction between PCl_3 and Cl_2 to form PCl_5 the units of Kc are.

(A) (B)

(C) (D)

Q6. The conjugate acid of HPO_4^{2-} is

(A) (B)

(C) (D)

Q7. General formula of alkyl radical is.

(A) (B)

(C) (D)

Q8. Coal gas is mixture of.

(A) (B)

(C) (D)

Q9. A hydro carbon has molecular formula of C_8H_{14}. What is the molecular formula of the next member of same homologous series

(A) (B)

(C) (D)

Q10. Night blindness is because of deficiency of

(A) (B)

(C) (D)

Q11. Percentage amount of Argon in Air.

(A) (B)

(C) (D)

Q12. Normally rain water is weakly acidic because of

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 1022 (جماعت دہم) — سیکنڈری پارٹ II — Session 2018-20 to 2020-22 — کیمسٹری (انشائیہ) — حصہ اول — گروپ پہلا

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Write the use of Oxygen and Nitrogen in the manufacture of chemicals? ii. What is irreversible reaction? Give example. iii. Define Molecular formula? Give Example. iv. Write two basic conditions for an element for catenation? v. What are Isomerism? Give example. vi. How is an Amino group tested? vii. What is meant by Tri-glycerides? Write their general formula? viii. Write the two properties of polysaccharides?

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Water is an amphoteric compound. Give an equation. ii. Write two uses of Acetic acid. iii. Why methane is called marsh gas? iv. Why alkenes are called olefins. v. Write composition and boiling range of Diesel oil. vi. Draw and label the diagram of Bessemer Converter. vii. Write any two raw materials for solvay's process. viii. What is Calcination? Give reaction.

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. What do you mean by atmosphere? ii. Identify as primary or secondary air pollutants. CH_4, NH_3, H_2SO_4, O_3 iii. How is Ozone layer formed in stratosphere? iv. What are pollutants and contaminants? v. How does water dissolve sugar and alcohol? vi. Mention two disadvantages of detergents. vii. Why are pesticides used? viii. What are waterborne infectious diseases?

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. Explain structure of DNA and DNA have genetic code of life. 3+2

b. How can we predict the extent of reaction by equilibrium constant? 1×4=4

Q6. 6.a. Define a salt and give the characteristic properties of salts. 1+4=5

b. Explain the reasons why water is considered a universal solvent. 2+2

Q7. 7.a. Explain the Bassemerization with balance chemical equations. 1+2+2

b. Write any four uses of Ethylene (Ethene) 1×4=4

Note: Board typos kept verbatim: 'Bassemerization', 'solvay's', 'Give Example.'.

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 1022 (**جماعت دہم**) — **سیکنڈری پارٹ** II — Session 2018-20 to 2020-22 — **کیمسٹری** (**معروضی**) — **گروپ دوسرا**

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 3486

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. About 99% of atmosphere's mass lies within

(A) (B)

(C) (D)

Q2. Permanent hardness is removed by adding

(A) (B)

(C) (D)

Q3. Which one of the following diseases causes liver inflammation

(A) (B)

(C) (D)

Q4. Crude oil is heated in furnace upto

(A) (B)

(C) (D)

Q5. At dynamic equilibrium

(A) (B)

(C) (D)

Q6. Which one of the salts is used to dry a gas?

(A) (B)

(C) (D)

Q7. The ability of carbon atoms to form chain is called

(A) (B)

(C) (D)

Q8. Which one of the following is a synthetic fibre?

(A) (B)

(C) (D)

Q9. Oxidation of ethene with $KMnO_4$ produces

(A) (B)

(C) (D)

Q10. Photosynthesis process produces

(A) (B)

(C) (D)

Q11. Normally rain water is weakly acidic due to the gas

(A) (B)

(C) (D)

Q12. Global warming causes rising of sea level, The cause of global warming is

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 1022 (**جماعت دہم**) — **سیکنڈری پارٹ** II — Session 2018-20 to 2020-22 — **کیمسٹری (انشائیہ) — حصہ اول — گروپ دوسرا**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Give two characteristics of reversible reactions. ii. What is relationship between active mass and rate of reaction? iii. Why are the melting and boiling points of organic compounds low? iv. Define aromatic compounds. Give an example. v. Why are the organic compounds

poor conductors of electricity? vi. Define organic compounds Give an example. vii. What is the difference between glucose and fructose? viii. What is the function of DNA?

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. How can you Justify that water is an Amphoteric compound? ii. Describe two uses of acetic Acid. iii. Justify that alkanes give substitution reaction. iv. Why does the burning of alkanes require sufficient supply of oxygen? v. What is difference between crude oil and residual oil? vi. What is the raw material for the manufacture of Urea? vii. Write two formulas of copper ores. viii. How slag is produced in smelting process?

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. How waterborne diseases can be prevented. ii. How water rises in plants. iii. State the major sources of CO and CO_2 emission. iv. How does acid rain increase the acidity of soil. v. What are the reasons of waterborne diseases. vi. How does limestone dissolve in water. vii. Explain the phenomenon of decreasing temperature in troposphere. viii. How is Ozone layer formed in stratosphere.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. What are Polysaccharides? Write any four uses of carbohydrates. 1+4

b. State the Law of mass action and derive the expression for Equilibrium Constant for a general reaction. 4

Q6. 6.a. Give uses of five different Bases in daily life. 1×5=5

b. Explain two Methods of removing Temporary hardness. 2+2=4

Q7. 7.a. Write the five advantages of Solvay's process? 1×5=5

b. Describe two Methods of prepration of Alkynes using chemical equation? 2+2

> Note: Board typos kept verbatim: 'prepration'.